

Hong Tang

Research Assistant Professor, Department of Building Environment and Energy Engineering
The Hong Kong Polytechnic University
honghong.tang@polyu.edu.hk | Google Scholar: nXW7KrwAAAAJ | ResearchGate: Hong-Tang-17

Profile

- Research Assistant Professor at The Hong Kong Polytechnic University. Research focuses on energy-flexible and grid-responsive buildings, demand response, renewable and storage optimization, and AI-enabled energy-system control.
- Published 20 papers; Google Scholar citations: 811.

Research Interests

Grid-interactive buildings; building energy flexibility and demand response; AI and machine learning for energy systems; data-center precision cooling and computing-power/electricity coordination; integrated energy systems and virtual power plants.

Education

- PhD, Building Environment and Energy Engineering, The Hong Kong Polytechnic University, 2018-2023.
- BEng, Building Environment and Energy Application Engineering, Tsinghua University, 2014-2018.

Appointments

- Research Assistant Professor, The Hong Kong Polytechnic University, 2025-present.
- Postdoctoral Fellow, The Hong Kong Polytechnic University, 2023-2025.
- Research Assistant, The Hong Kong Polytechnic University, 2022-2023.

Selected Publications

- Hong Tang et al. Multi-Timescale Rolling Optimization Coordinating Heterogeneous EV and Building Load Flexibility in Community Energy Systems with NGBoost-Based Renewable Forecast. *Applied Energy*, accepted, 2026.
- Hong Tang et al. Risk-aware optimal dispatch of resource aggregators integrating NGBoost-based probabilistic renewable prediction and bi-level building flexibility engagement. *Engineering*, 53:76-89, 2025.
- Hong Tang et al. Unlocking Building Energy Flexibility: Multi-Layered Pricing Framework Design from Distribution-Level to P2P Markets. *The Innovation Energy Use*, 1(2):100019, 2025.
- Hong Tang and Shengwei Wang. Game-theoretic optimization of demand-side flexibility engagement considering stakeholder perspectives and multiple flexibility services. *Applied Energy*, 332:120550, 2023.
- Hong Tang, Shengwei Wang and Hangxin Li. Flexibility categorization, sources, capabilities and technologies for energy-flexible and grid-responsive buildings. *Energy*, 219:119598, 2021.

Service and Awards

- Young Editorial Board Member, *The Innovation Energy Use*.
- Guest Editor for *Energy*, *Building Simulation*, *Buildings*, *Architecture*, *Journal of Physics: Conference Series*, and *Journal of Building Design and Environment*.
- Best Presentation Award, The 11th Asia Conference on Power and Electrical Engineering, 2026.
- Best Poster Award, The 4th Asia Conference of International Building Performance Simulation Association, 2018.

Selected Projects

- PI, PolyU Start-up Fund: Large-scale Flexible Control and Supply-demand Coordinated Optimization of Integrated Energy Systems towards Carbon Neutrality, HK\$300,000, 2025-2027.
- PI, PolyU Postdoctoral Matching Fund: Study on systematic quantification method, design and control optimization of energy-efficient and grid-responsive buildings, HK\$365,000, 2023-2025.
- Key researcher, China Southern Power Grid project on model-data hybrid rapid evaluation of building energy flexibility, RMB 2.3867 million, 2024-2026.

Last updated: May 27, 2026